



THE UNIVERSITY OF LARKANO

DEPARTMENT OF COMPUTER SCIENCE



Course Title	Object Oriented Programming
Course Code	CC-312
Department	Computer Science
Batch	24(F)
Semester and Year	2nd Semester, 1st Year
Instructor Name	Soban Hussain

Course File index

S.#	Document	Included
01	Syllabus (Mentioning CLO, Taxonomy Levels, CLO Mapping with PLO and reference books)	✓
02	Tentative Teaching Plan	✓
03	Class Timetable	✓
04	Grading Policy & Activities Weightage	✓
05	Midterm and Final Exam Schedule	✓
06	Quiz, Assignments, Midterm and Final Exam Question Papers	✓
07	OBE Based Result	✓
08	Course Analysis Form	✓
09	Best, Average and Worst Answer Sheets by the Students (Test/Quiz, Assignment, Mid Exam)	✓
10	Model Solutions for the Assessment Tools (Test/Quiz, Assignment, Mid Exam, Final Exam)	✓



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COURSE TITLE: OBJECT ORIENTED PROGRAMMING	COURSE CODE: CC-312	CREDIT HOURS: 03	MIN CONTACT HRS: 32
COURSE INSTRUCTOR: SOBAN HUSSAIN			
Batch: 24(F)	Semester: 2nd Semester, 1st Year	Year: 1st	Semester Commenced On:
Course Learning Outcomes			
CLO #	Description	Learning Domain	Taxonomy Level / PLO #
1	Understand principles of object-oriented paradigm and java	Cognitive	Level C2 / PLO 2
2	Identify the objects & their relationships to build object-oriented solution	Cognitive	Level C3 / PLO 3
3	Model a solution for a given problem using object oriented principles	Cognitive	Level C3 / PLO 4
Lesson Contents with Associated Details			
Course Contents	CLO # / Marks	Delivery Techniques	Assessment Techniques (Marks)
• Introduction to Object Oriented Programming • OOP Concepts • Basic Programming in Java • Classes and Objects • Access Modifiers	CLO 1 40 Marks	Class Lectures Tutorials Total Classes: 6	• Assign (5) • Test (5) • Mid (10) • Final (20)
• Syntax, Datatypes, Arrays, Strings • Operators • Loops • Method & Functions • Static Members & Functions • Function & Operator Overloading • Class Relationships • Composition & Aggregation • Classes & Objects	CLO 2 30 Marks	Class Lectures Tutorials Total Classes: 8	• Assign (3) • Test (3) • Mid (5) • Final (19)

• Data Encapsulation • Constructor & Destructor • Inheritance & Polymorphism • Abstract Classes & Interfaces			
• Generic Programming • Templates & Standard Template Library (STL) • Object Streams & Serialization • Exception Handling	CLO 3 30 Marks	Class Lectures Tutorials Total Classes: 14	• Assign (2) • Test (2) • Mid (5) • Final (21)

Assessment Technique / Activity	Marks Assigned	No. of Activities	CLO(s) Measured
Class Test / Quiz	10	2	1, 2
Assignment	10	1	1
Mid Semester Exam	20	1	1, 2
Final Semester Exam	60	1	1, 2, 3



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Tentative Teaching Plan

Week	Topic	Activity
01	Introduction to Object Oriented Programming	
02	Programming in Java (Foundation: Datatypes, Input, Arrays, String, loops, User-defined methods)	Assignment
03	OOP Concepts, Classes & Objects	
04	Data Encapsulation, Constructors & Destructors	Test #1
05	Function & Operator Overloading, Access Modifiers, Static Members & Functions	
06	Class Relationships, Composition & Aggregation	
07	Inheritance & Polymorphism	
08	Mid Term Exam	
09	Abstract Classes & Interfaces	
10	Generic Programming	
11	Function & Class Templates, STL	
12	Object Streams & Serialization	Test #2
13	Exception Handling	

14	Exception Handling	
15	Course Revision	
	Final Exam	



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Course Teacher: Soban Hussain

Semester and Year: 2nd Semester, 1st Year

A. Assessment Tools and Grading Policy

Theory (100 Marks)	Assessment Tools		GPA Weight
	Mid Term Exam		20%
	End Semester Exam		60%
	Sessional	Class Tests / Quiz	10%
		Assignments	10%
	Total		100%

B. OBE Weightage Set by Teacher

CLO #	Sessional Weight		Mid-Term Exam Weight	Final Term Exam Weight	Total Weight Assigned to a CLO
	Assignment	Tests			
CLO 1	5%	5%	Q.1(a)+Q1(b) (10%)	Q.1 +Q.2 (24%)	40%
CLO 2	3%	3%	Q.2(a)+Q2(b) (10%)	Q.3(12%)	30%
CLO 3	2%	2%		Q.4+Q5(24%)	30%
Total Weight Assigned to A.T.	20%		20%	60%	100%